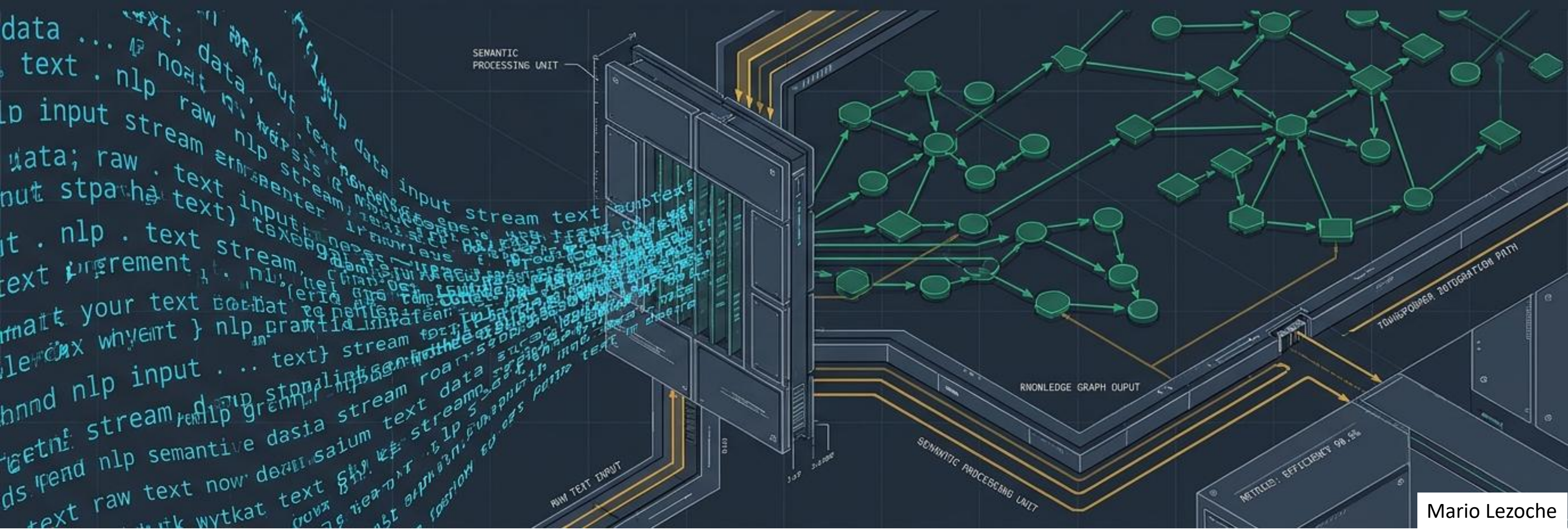


Ontology-Driven NLP, LLM Data Synthesis, and Transformer Benchmarking

Ontology-Driven NLP, LLM Data Synthesis, and Transformer Benchmarking



The Engineering Mandate

Design an end-to-end software system that transforms unstructured text into structured RDF triples, governed by a custom ontology and benchmarked against multiple ML architectures.



1. Domain Ontology

Formalize concepts, relations, and explicit semantic constraints.



2. Synthetic Dataset

Utilize LLMs to generate tiered, annotated text data (JSONL).



3. Extraction Pipeline

Build modular NLP software to detect entities and extract relations.

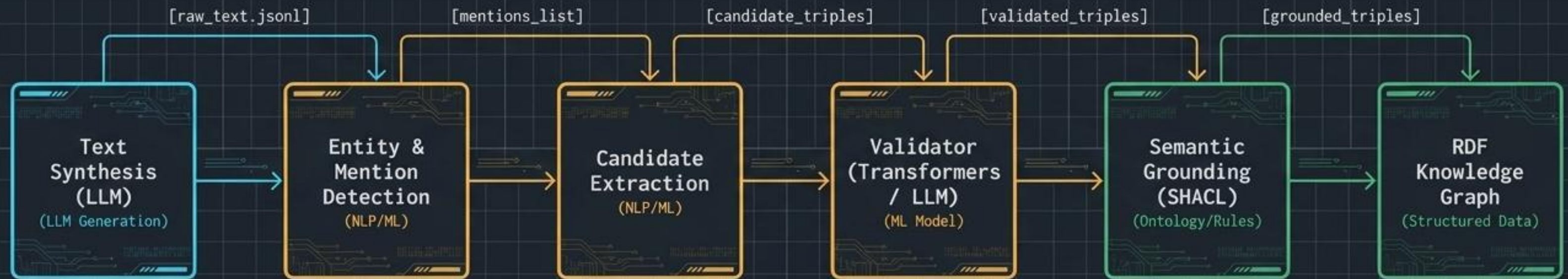


4. Model Benchmarking

Rigorously evaluate multiple Transformer architectures (SpaCy baseline vs. BERT/LLM validators).

The Data Refinery Pipeline

An end-to-end schematic for unstructured-to-structured data transformation.

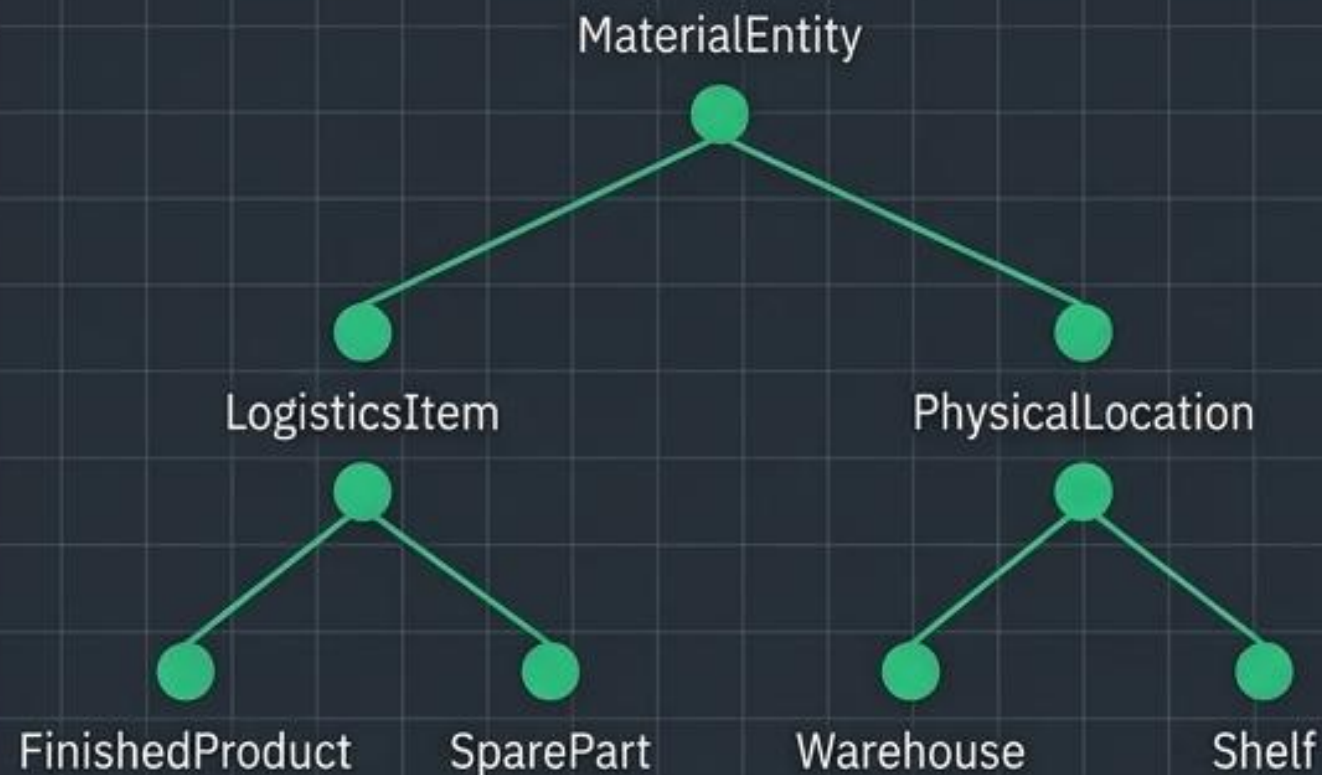


Phase 1: Semantic Foundation

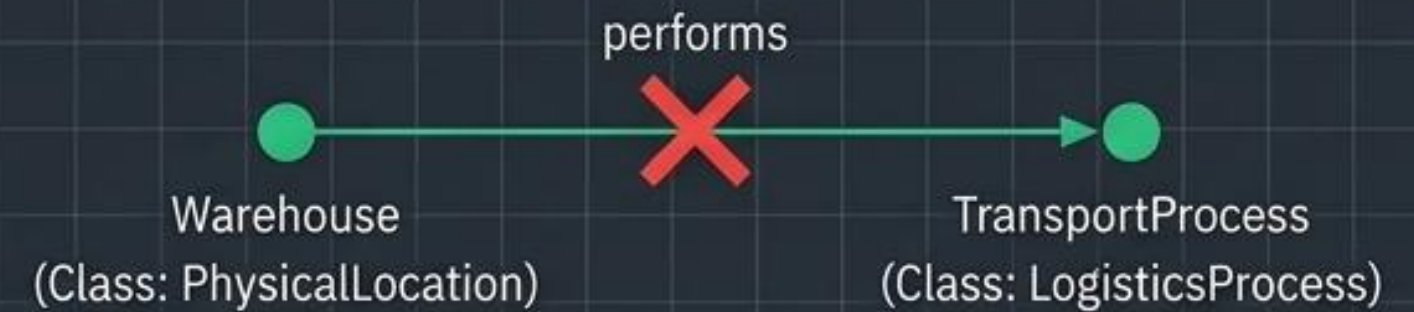
Define formal concepts, directional relations (forward/inverse), and explicit constraints (domain/range). The ontology does not generate data; it restricts and validates it.



Hierarchical Classes



Constraint Gate (Domain/Range)



ERROR: Relation 'performs' Domain is strictly constrained to 'Agent'. 'PhysicalLocation' is invalid.

Phase 2: Synthetic Data Generation via LLMs

Requirement: 70/15/15 Data Split (Train / Validation / Test)



Tiered Complexity Requirements

1. Explicit

Direct trigger words in a single, simple sentence.

2. Implicit

Requires synonym resolution or broader context.

3. Long-Distance

Entities separated by multiple clauses or sentences.

4. Nested Relations

Dense, ambiguous text with multiple overlapping triples.

Phase 3: Text Preprocessing & Mention Detection



Raw Sentence

The server at 192.168.1.100 experienced high latency.

Tokenization

'The', 'server', 'at', '192.168.1.100', 'experienced', 'high', 'latency', '.'

Lemmatization

'The', 'server', 'at', '192.168.1.100', 'experience', 'high', 'latency', '.'

Entity Linking (IRIs)

'server' -> <http://data.refinery/ont#Server>
'192.168.1.100' -> <http://data.refinery/ont#IPAddress>
'high latency' -> <http://data.refinery/ont#HighLatencyEvent>

Entity Detection Implementation Choices	
Classical Rule-Based	Label-Guided NER (GLiNER)
Method: Dictionary matching from OWL labels .	Method: Detects entity types based on semantic descriptions.
Mechanic: Longest string match first.	Mechanic: Dynamic semantic roles derived from the ontology .
Trade-off: High precision, zero adaptability to new vocabulary.	Trade-off: Highly adaptable, requires compute for embedding generation.

Phase 4: Relation Extraction (The SpaCy Baseline)



Baseline Limitations

- Brittle on complex or nested sentences.
- Heavy trigger-based bias.
- Rapidly dropping recall on implicit relations.

Bulbasaur is a Grass and Poison type, at least in his notes.



ACTUAL TRIPLES:

```
('http://example.org/pokemon#species_0001', 'http://example.org/pokemon#hasType', 'http://example.org/pokemon#Grass')
('http://example.org/pokemon#species_0001', 'http://example.org/pokemon#hasType', 'http://example.org/pokemon#Poison')
```

PREDICTED TRIPLES:

```
('http://example.org/pokemon#species_0001', 'http://example.org/pokemon#hasType', 'http://example.org/pokemon#Grass')
('http://example.org/pokemon#species_0001', 'http://example.org/pokemon#hasType', 'http://example.org/pokemon#Poison')
```

OK

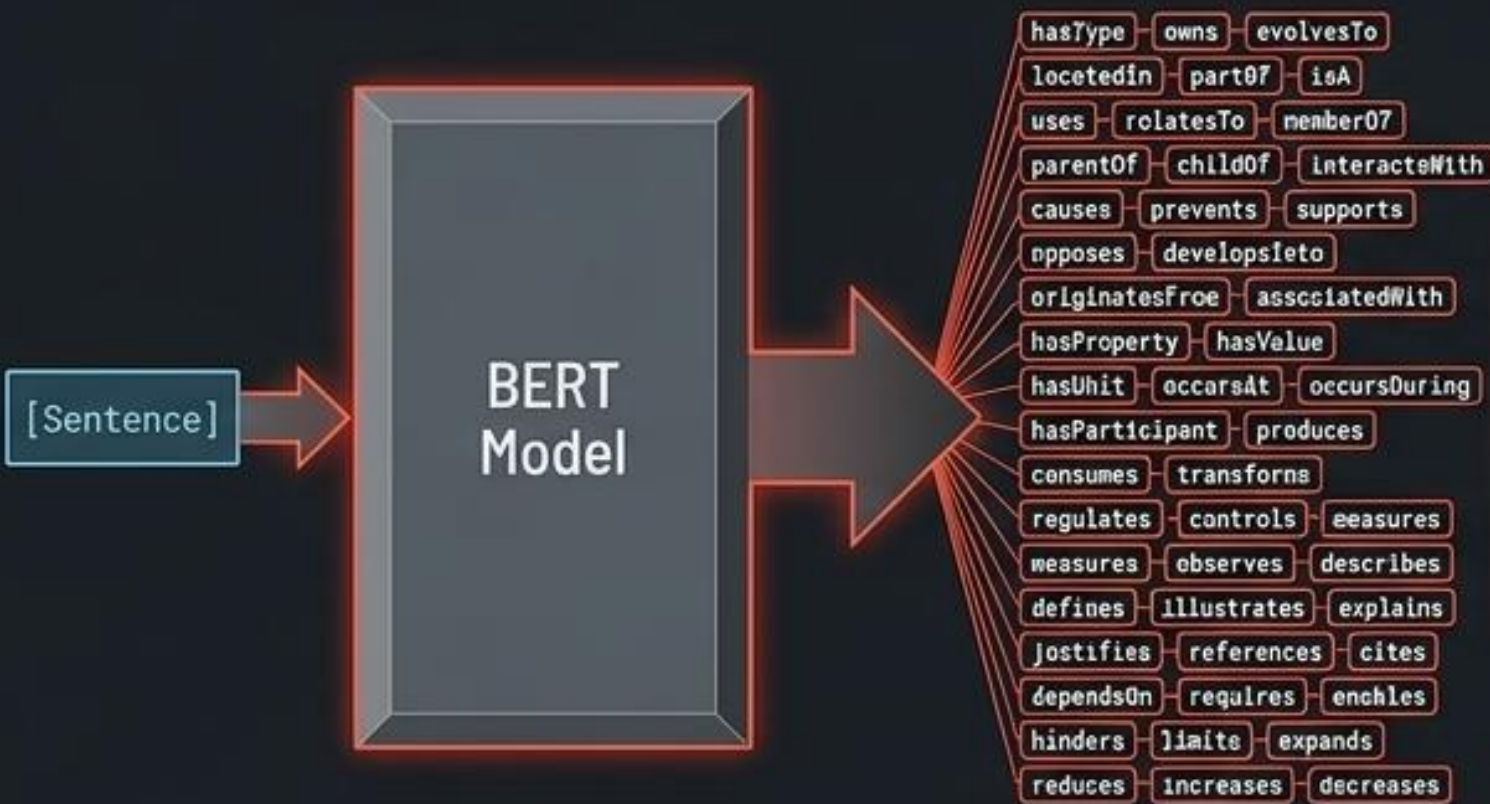
The Data Refinery



The Architectural Shift: From Generator to Validator

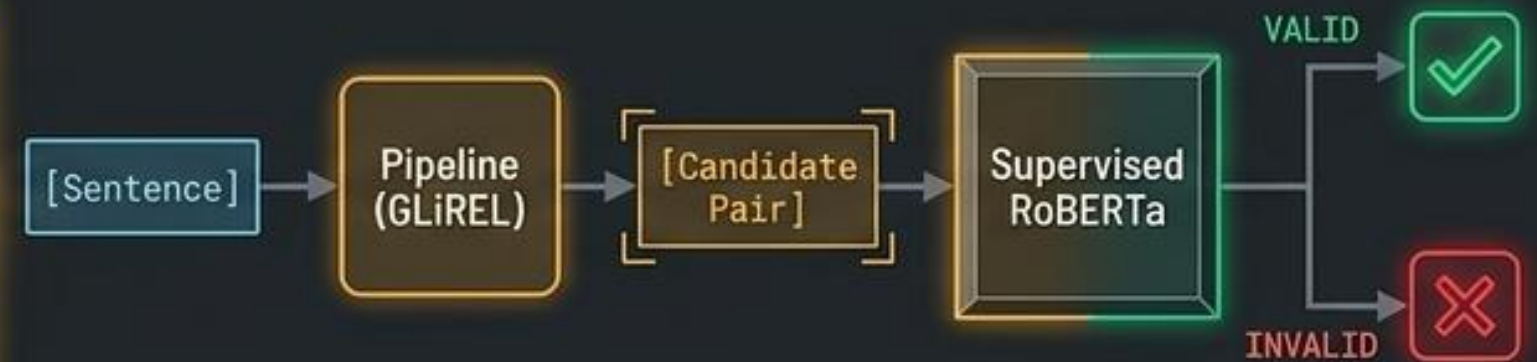
Traditional BERT extraction creates an 'ontology-compiled model' that breaks when semantics change. The solution is to dissociate candidate generation from supervised decision-making.

The Old Way (Generator)



Model must guess the exact relation from scratch. Brittle to ontology updates.

The New Way (Validator)



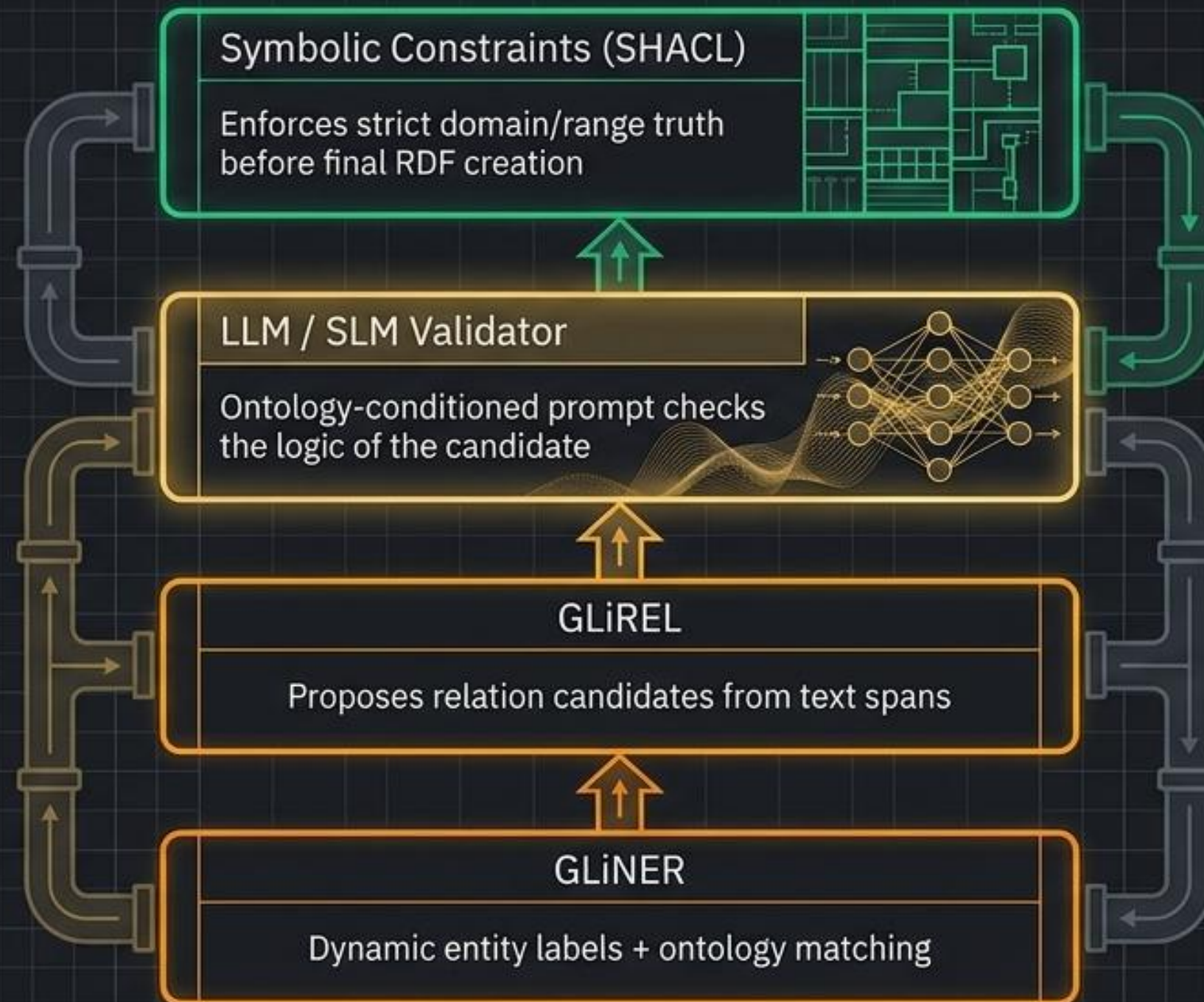
Model only acts as a stable logic gate for proposed relations.

Phase 5: Advanced Architectural Paradigms

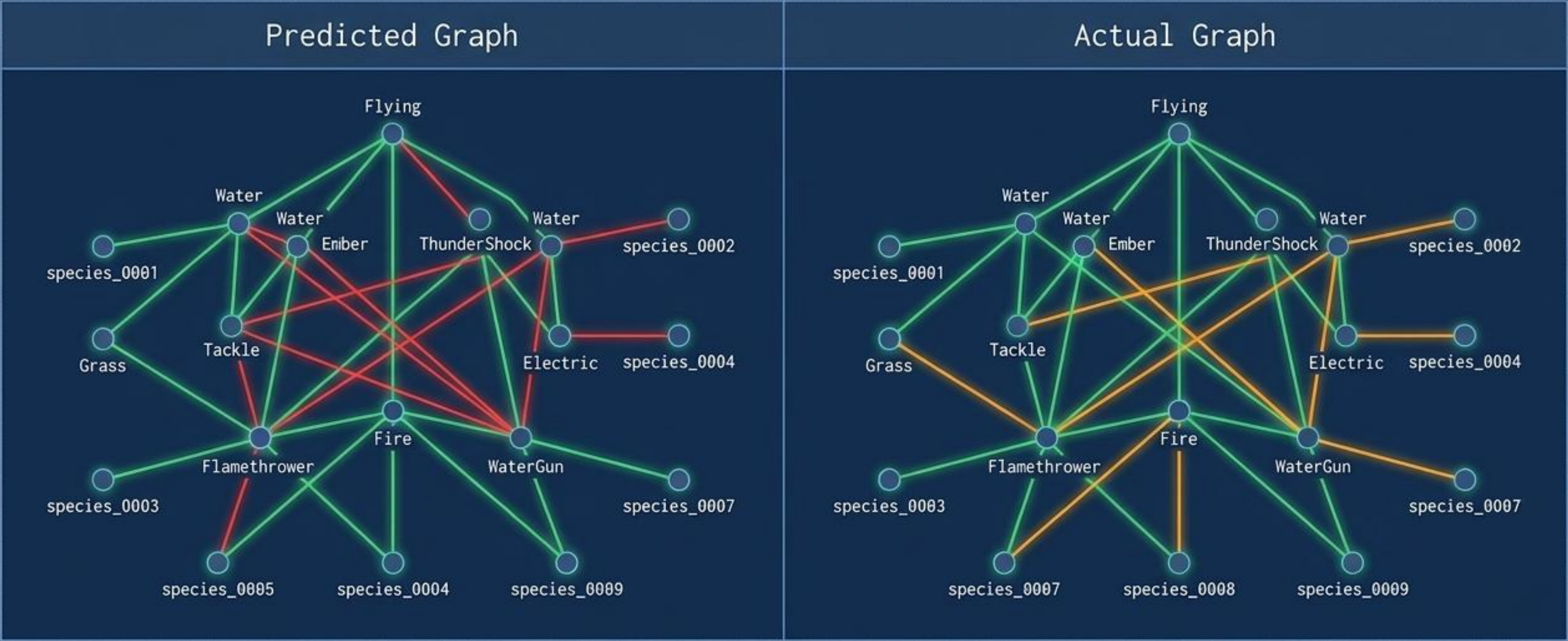


Architecture A: Schema + Verifier	Architecture B: Abstract Mapping	Architecture C: Dynamic LLM Validator
1 GLiNER / GLiREL candidate generation.	1 Pairwise classification into abstract relations.	1 GLiNER entity extraction.
2 Supervised BERT binary validator.	2 Rule-based/BERT ontology mapping.	2 SLM / LLM prompt validates semantics.
		3 SHACL enforces truth.
Trait: Stable learning, completely independent of concrete ontology changes.	Trait: Intermediate approach. Requires mapping table updates.	Trait: Highly dynamic and adaptable, but computationally expensive.

Deep Dive: Dynamic LLM Validator Pipeline



Phase 6: Knowledge Graph Assembly



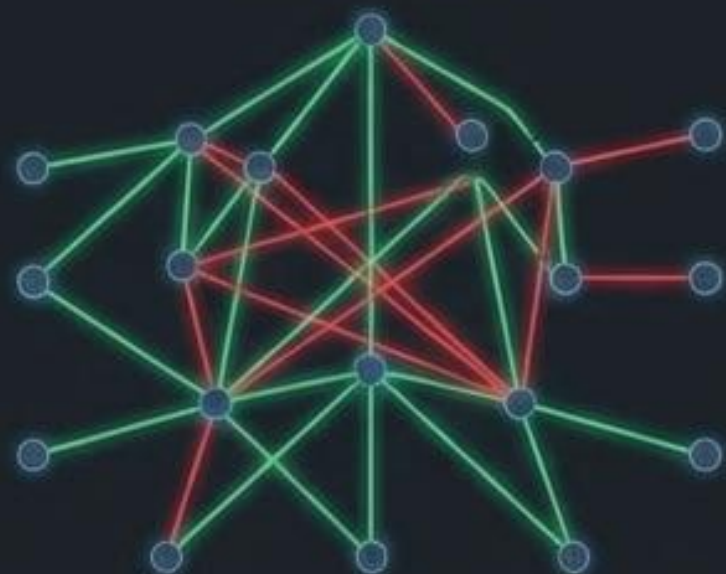
[—] True Positive | [—] False Positive | [—] False Negative

The Evaluation Framework



Precision: $TP / (TP + FP)$

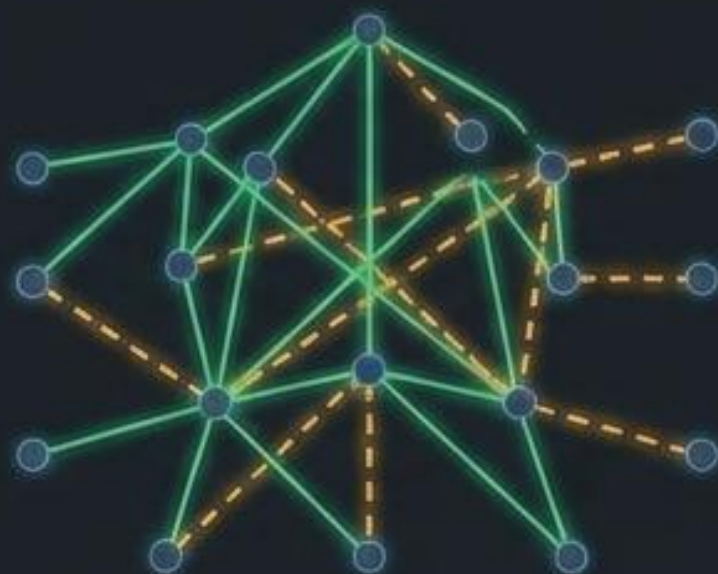
Predicted Relations



How many predicted relations are actually correct?

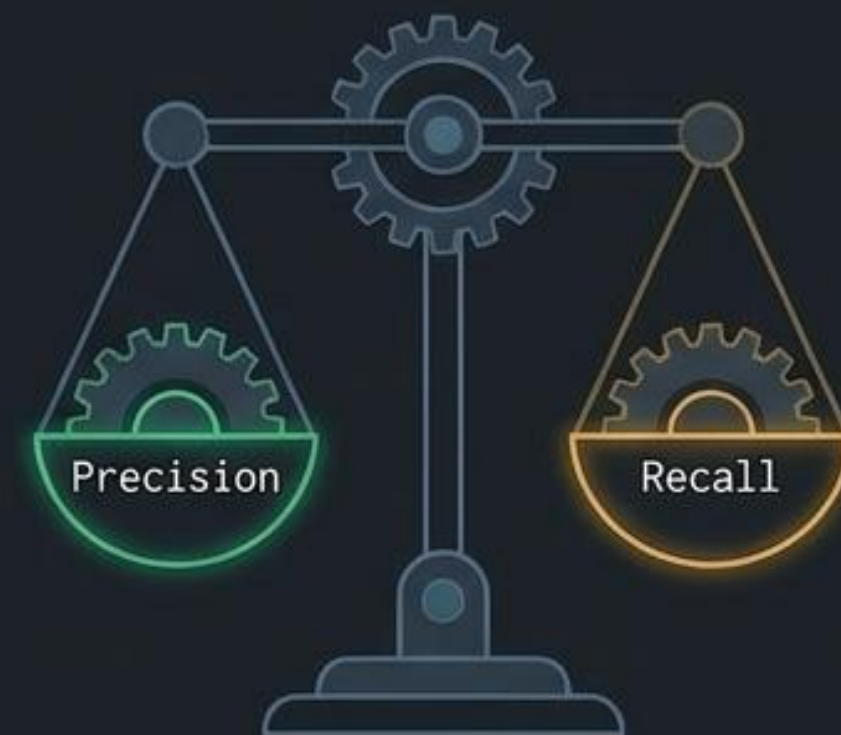
Recall: $TP / (TP + FN)$

Actual Ground-Truth Graph



How many actual relations did the system successfully recover?

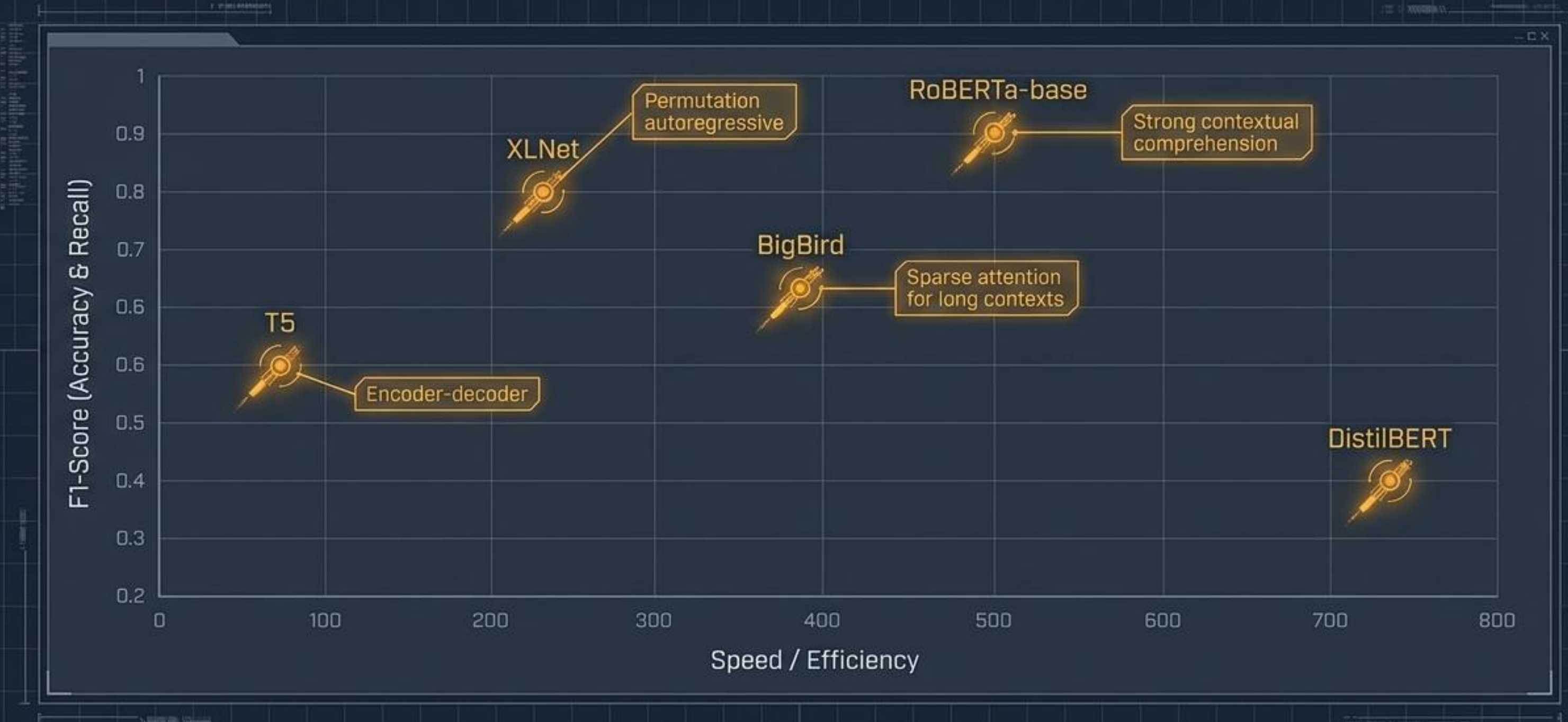
F1-Score



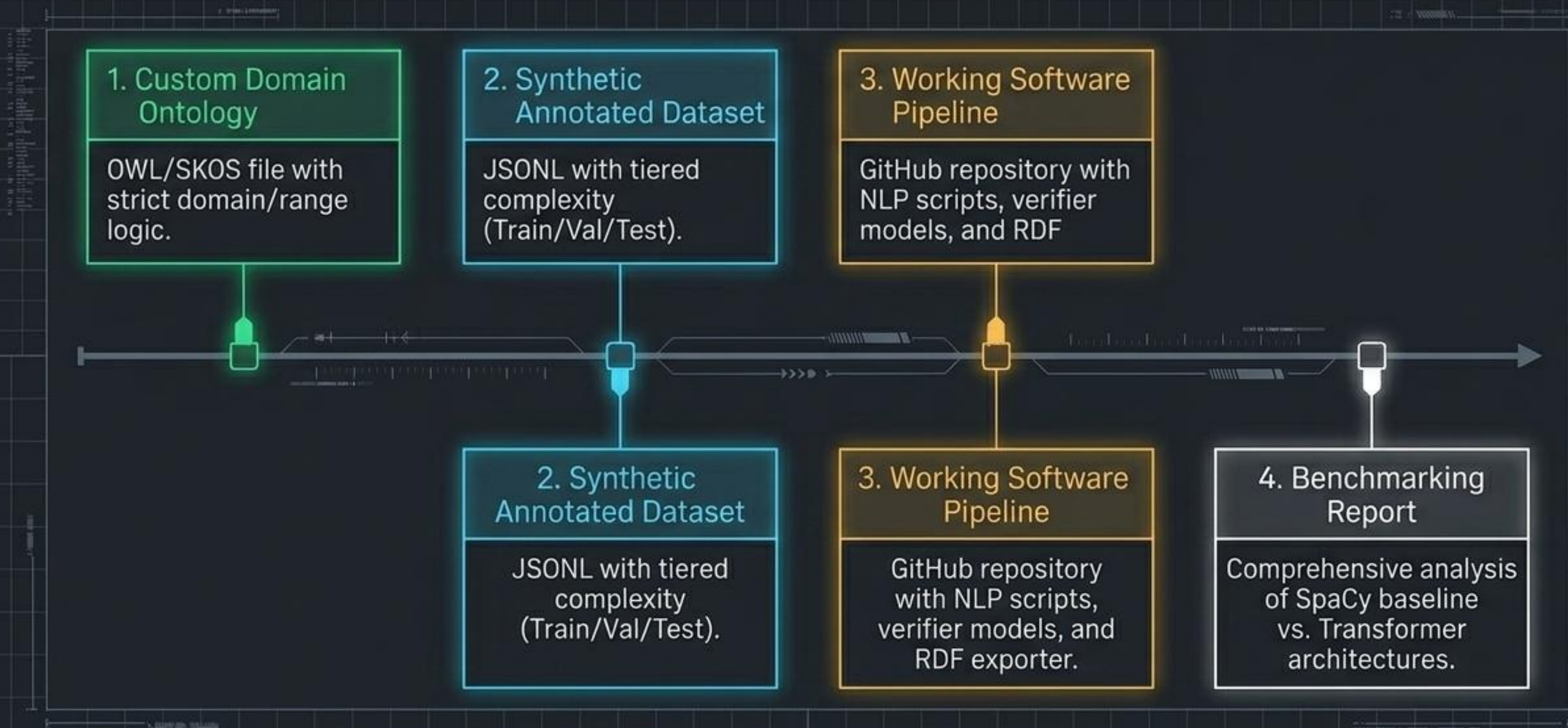
The harmonic mean balancing extraction accuracy and coverage.

Benchmarking Requirements

Analyze architectural trade-offs across Transformer families.



Deliverables & Milestones



Building Systems That Understand the World

You are not just parsing text. You are designing deterministic boundaries around probabilistic models to build reliable, human-centric intelligent enterprise systems.

